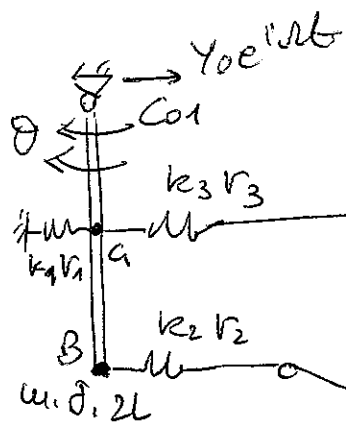
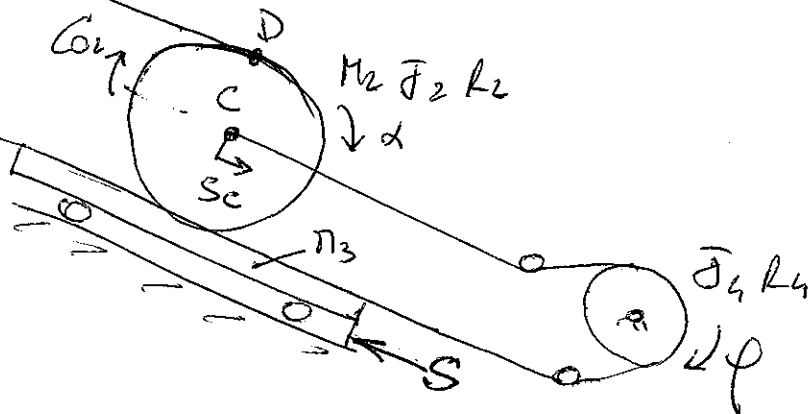


2 dic 2010 - B



$$\underline{x} = \begin{bmatrix} \theta \\ \varphi \\ y \end{bmatrix}^T$$



$$E_c = \frac{1}{2} m_1 \dot{x}_A^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 v_C^2 + \frac{1}{2} J_2 \dot{\alpha}^2 + \frac{1}{2} m_3 v^2 + \frac{1}{2} J_4 \dot{\varphi}^2$$

$$x_A = -L\theta + y$$

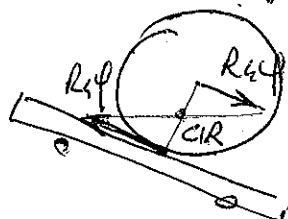
$$S = R_4 \varphi$$

$$S_C = R_4 \varphi$$

$$\alpha = \frac{2 R_4 \varphi}{R_2}$$

$$\underline{y}_{mf} = \begin{bmatrix} x_A & \theta & S_C & \alpha & S & \varphi \end{bmatrix}^T$$

$$[M_F] = \text{diag}(m_1, J_1, m_2, J_2, m_3, J_4)$$



$2 R_4 \varphi = \text{spost. relativo}$
centro disco - canello

$$[A_m] = \begin{bmatrix} -L & 0 & 1 \\ 1 & 0 & 0 \\ 0 & R_4 & 0 \\ 0 & 2R_4/R_2 & 0 \\ 0 & R_4 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{cases} \Delta L_1 = y - L\theta \\ \Delta L_2 = -S - x_B \\ \Delta L_3 = -x_A + x_D \\ x_D = S_C + R_2 \alpha \end{cases}$$

$$U_{el} = \frac{1}{2} k_1 \Delta L_1^2 + \frac{1}{2} k_2 \Delta L_2^2 + \frac{1}{2} k_3 \Delta L_3^2$$

$$[A_{el}] = \begin{bmatrix} -L & 0 & 1 \\ 2L & -R_4 & -1 \\ L & 3R_4 & -1 \end{bmatrix}$$

$$\begin{cases} \Delta L_1 = y - L\theta \\ \Delta L_2 = -y + 2L\theta - R_4 \varphi \\ \Delta L_3 = -y + L\theta + 3R_4 \varphi \end{cases}$$

$$V_1 = -m_1 g L \cos \theta \rightarrow \frac{\partial^2 V_1}{\partial \theta^2} = m_1 g L \cos \theta \quad [K_1] = \begin{bmatrix} m_1 g L & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\delta L = C_1 \delta \theta + C_2 \delta \alpha + R_x \delta y =$$

$$= C_0 \delta \theta + C_0 \frac{2R_1}{R_2} \delta \varphi + R_2 \delta \gamma$$

$$Q_{c1} = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix} C_{01} e^{1st}$$

$$Q_{C2} = \begin{Bmatrix} 0 \\ 2k_2 \\ k_2 \\ 0 \end{Bmatrix} \text{ Core ist}$$

$$\underline{Q_v} = \{ R_x \}$$

CALCOLO REAZIONE VINCOLE
CON EQUILIBRI DINAMICI

$m_1 L \dot{\theta}^2$: Termine dell'azione

$$1) \sum F_x = 0: R_x - \bar{F}_{el1} + \bar{F}_{el2} + \bar{F}_{el3} + \omega_1 L \dot{\theta} = 0$$

$$\rightarrow h_x = + \text{Fe}^{+1} - (\text{Fe}^{+2} + \text{Fe}^{+3} + w_1 \text{LO}^{\cdot\cdot})$$

$$\bar{F}_{el1} = (k_1 + 1.4v_1) \Delta L_1 \quad ; \quad \Delta L_1 = -0.0$$

$$F_{\ell 2} = (k_2 + i\alpha k_2) \Delta \ell_2 \quad ; \quad \Delta \ell_2 = 2L\theta - R_4\varphi$$

$$F_{el3} = (k_3 + i c_3 \dot{r}_3) \Delta l_3; \quad \Delta l_3 = L\theta + 3R_4\varphi$$

$$\ddot{\theta} = -\omega^2 \theta$$

$$2) \sum M_i = 0 : C_1 - T_1 \hat{\theta} - R_y L \theta + R_x L - F_{el2} L = 0$$

$$\sum F_y = 0 : R_y - m_1 g = 0 \rightarrow R_y = m_1 g$$

$$\rightarrow R_x = (F_{el2} L + m_1 g L \theta + I_1 \ddot{\theta} - G_1)$$

L

